



Artificial intelligence of things (AIoT) for precision agriculture: applications in smart irrigation, nutrient and disease management

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ABSTRACT

Despite the growing global demand for efficient and sustainable agricultural practices, many traditional farming systems still rely on manual decision-making, which can be inefficient and resource intensive. This review comprehensively examines the applications of Artificial Intelligence of Things (AIoT) in three critical domains of precision agriculture, such as smart irrigation, nutrients and disease management. Through the deployment of interconnected sensor networks, edge and cloud computing platforms, and AI algorithms, AIoT systems facilitate site-specific monitoring and control key environmental and crop parameters. In irrigation, AIoT empowers farmers to optimize water distribution through real-time soil moisture sensing, predictive analytics and dynamic irrigation scheduling. Precision nutrient management utilizes unmanned aerial vehicles, soil sensors, and AI-powered data analysis to monitor nutrient availability and inform optimized fertilization strategies, thereby improving nutrients efficiency and reducing environmental degradation. Similarly, AIoT in disease management enhances surveillance and predictive capabilities by integrating sensor data with AI models to early detect abiotic stresses, enabling timely and targeted interventions. Although the significant benefits of AIoT, its implementation faces several challenges. These include high setup costs, data connectivity issues in rural areas, inconsistent sensor reliability, cybersecurity risks, and limited scalability for smallholder farming systems. Additionally, gaps in technical knowledge and infrastructure pose barriers to widespread adoption. However, advancements in 5G technology, edge computing, and sensor miniaturization offer promising avenues for scaling AIoT in agriculture. This review therefore highlights the transformative potential of AIoT in improving climate-smart and sustainable agriculture by converting data into actionable insights to enhance resilience and food security.

1. Introduction

Agriculture, the backbone of global food security and rural economies, is undergoing a revolutionary transformation driven by cutting edge technologies. Among these, the convergence of AI and IoT collectively referred as AIoT has become increasingly significant in the field of smart agriculture [1]. AIoT integrates the real time data collection capabilities of IoT devices with data analysis and decision-making powers of AI, creating an intelligent ecosystem that enhances agricultural productivity, resource efficiency, and environmental sustainability [2]. Conventional farming is no longer only dependent on manual labor and traditional knowledge. Instead, it is increasingly dependent on intelligent systems that can predict crop diseases, monitor soil and plant health, automate irrigation systems, and optimize nutrient applications.

Smart agriculture and dynamic farming techniques have been examined and integrated into traditional farming systems to improve agricultural productivity [3]. The IoT has recently transformed agricultural practices, leading to the emergence of smart agriculture [4–8]. AIoT combined with robotics has the potential to make commercial farms more sophisticated, while reducing the need for manual labor from 90 % to 10 % [9]. AIoT enables farmers to make timely and informed decisions, reduce resources wastage, and respond proactively to threats such as drought, nutrient deficiencies, and pest infestations. It enables users to operate and manage agriculture in a more accessible, efficient, and independent manner [10]. Its applications include sensor-based monitoring systems, satellite imagery, Unmanned Aerial Vehicles (UAV), weather stations, and Machine Learning (ML) models that help to predict and control complex farm variables. These technologies are

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particularly useful in addressing major challenges such as climate variability, water scarcity, fertilizer overuse, and chemical pesticide resistance which offer new ways to increase productivity while protecting the environment. IoT-based sensors continuously monitor and transmit key data on soil texture, moisture content, nutrients level, and temperature to a centralized cloud platform, where the ML algorithms analyze the information to generate actionable insights [11].

AIoT application in agriculture has rapidly expanded over the past decade, especially in technologically advanced and semi-advanced agricultural economies. Farmers, agronomists, and researchers are increasingly utilizing AIoT systems for soil monitoring, irrigation decisions, nutrient management, crop pests and disease control (Fig. 1). Water is a critical but increasingly scarce resource in global agriculture. Traditional irrigation methods, such as flood and furrow irrigation, often result in significant water losses due to inefficient water distribution, evaporation, runoff and deep percolation [12]. AIoT based smart irrigation systems address these challenges by using soil moisture sensors, weather forecasting models, satellite imagery, and crop growth data to determine the exact amount and time of water needed. For example, AI algorithms can analyze sensor data and weather predictions to adjust irrigation schedules dynamically. Integrating AI technology in smart greenhouses enhances crop yields, improves efficiency of fertilizer, and strengthens disease management through robotic systems and intelligent farming practices. Farmers can also utilize IoT technologies and affordable sensors to track light intensity, temperature, relative humidity, leaf wetness, and water flow on their farms [3,13–15]. IoT applications in smart farming uses various sensors to collect real time environmental data for controlling sprinkler systems which is a significant contribution in the field [16,17]. The IoT has been utilized to improve the efficiency of fertilizers and irrigation systems [18], while AI further improves forecasting and automation [19]. IoT-enabled valves and pumps carries out irrigation tasks without human intervention, ensuring crops receive optimal moisture levels [20], which not only conserves water but also improves plant health and yield.

Efficient nutrient management is vital for maximizing crop productivity and maintaining soil health. However, conventional fertilizer application methods often result in nutrient imbalances, runoff, leaching and environmental pollution. AIoT systems help to optimize nutrient application through real-time soil nutrient sensing, remote sensing

technologies, and ML models that recommend site specific fertilizer application rate and timing. Soil monitoring is an important component in agricultural systems [21], and AIoT system can integrate data from soil tests, crop growth models, and historical yield records to create a prescription map for variable rate application of nutrients. By offering a report on soil quality, these technologies assist farmers in applying fertilizer with the prediction accuracy of 98.75 % [22]. Pests and plant diseases are one of the major threats to food security and economic stability in agriculture. Conventional pest control methods often rely on broad spectrum chemicals, leading to pesticide resistance and environmental degradation. AIoT presents a more targeted and sustainable approach by integrating sensor networks, image recognition, remote sensing, and data analytics to monitor crop health in real-time [23]. AI models trained on large datasets can identify plant stress symptoms from images or spectral data and even distinguish between different pest species and diseases. Similarly, IoT-based spraying robots are developed that benefit agriculture in multiple ways including enhanced precision, improved effectiveness, and greater sustainability in pesticide application [24].

The main objectives of this review paper are:

- To examine the current state of AIoT applications in precision agriculture, particularly focusing on how combined AI and IoT systems improve the efficiency of smart irrigation, nutrient management, and disease control.
- To analyze various AI algorithms and IoT-based sensing and communication technologies used in real-time agricultural monitoring and automated decision-making.
- To highlight the environmental, economic and technical challenges limiting the widespread adoption of AIoT technologies in agricultural systems.

1.2. Novelty of the study

Numerous reviews [25–29], have explored the integration of AI and IoT in agriculture, most focus on wide technological backgrounds or single-domain applications. The innovation of the present study lies in its integrated, system level perspective of AIoT-driven precision

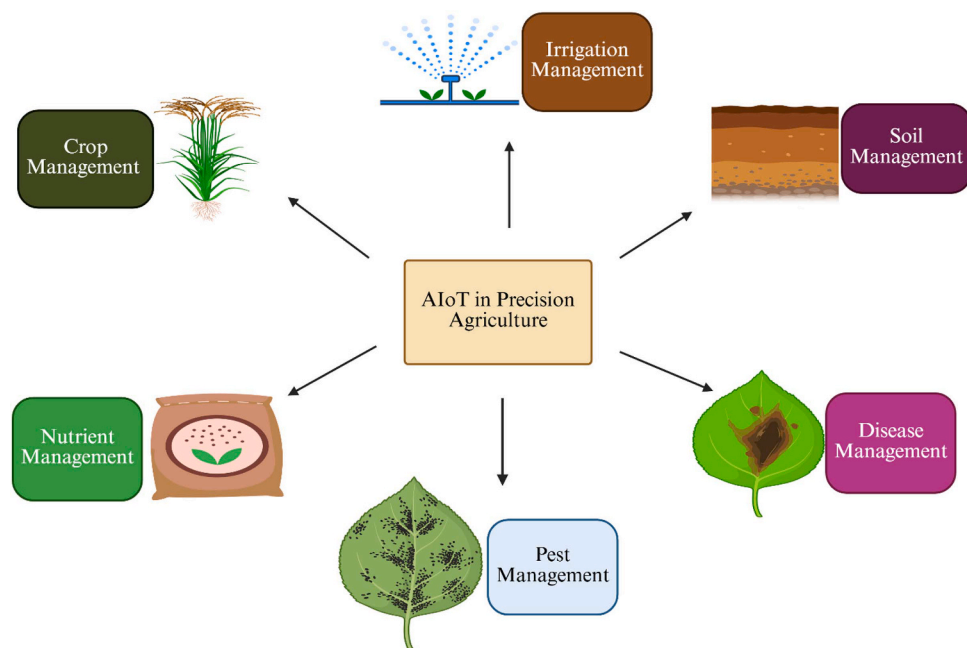


Fig. 1. Role of AIoT in enhancing core agricultural practices including soil, irrigation, nutrient and crop disease management.

agriculture, emphasizing real-time decision-making, multi-sensor data fusion, adaptive control systems across irrigation, nutrient, and pest and disease management. This review is also novel by synthesizing insights from peer-reviewed literature and real-world implementations, offering researchers, policymakers, and agricultural practitioners a comprehensive understanding of how AIoT is transforming the landscape of modern agriculture.

This review article is divided into five sections. The first section introduces the importance objectives of the study, followed by section discusses the applications of AIoT in precision agriculture, with a specific focus on its role in smart irrigation, nutrient and disease management. It also highlights recent developments and case studies that demonstrate the successful integration of AIoT systems across various agricultural practices. Furthermore, this section analyzes the key technological components of AIoT such as sensors, cloud computing, AI algorithms, and edge devices and explores how these elements interact within the broader precision agriculture ecosystem. The benefits of AIoT adoption, including enhanced crop yields, improved water and nutrient use efficiency, reduced operational costs, and greater sustainability, are also explored in detail. The next section identifies major challenges and limitations that hinder widespread adoption, such as data privacy concerns, high initial investment costs, limited technical training among farmers, and infrastructural constraints in developing regions. The final section discusses the opportunities, future directions, conclusions, and key takeaways of the review. Details of agriculture management practices and AIoT based solutions in precision agriculture are shown in Table 1.

2. AIoT for precision irrigation management

The integration of AIoT into irrigation systems has transformed how water resources are managed in agriculture. Traditionally, irrigation practices relied heavily on scheduling, manual observations, and farmer perception. With AIoT, this process is transformed into a real-time, intelligent operation. This approach significantly reduces water usage while improving crop health and yield. IoT-based smart irrigation systems can autonomously irrigate fields based on soil moisture conditions and weather predictions, achieving up to 80 % accuracy in controlled settings [30]. By optimizing water utilization, AIoT-driven irrigation addresses water scarcity concerns and contributes to sustainable agriculture practices [31]. Fig. 2 illustrates the application of AIoT based smart irrigation technologies in agriculture for water management.

2.1. Machine learning and AI models for predictive irrigation scheduling

ML and AI are the emerging powerful tools for predictive irrigation scheduling to optimize water use based on real-time field and historical

data [32]. Diverse datasets including weather attributes, soil moisture data, evapotranspiration, crop species and growth stages are integrated by different models to forecast when and how much to irrigate with accuracy [33]. ML models such as Random Forest (RF), Support Vector Machine (SVM), Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) algorithms are used widely to model the complex non-linear relationship in agricultural settings [34]. Integration of real time data from sensors with environmental variables, AI models can operate site specific irrigation decisions on enhanced water use efficiency by preventing over and/or under irrigation [35]. AIoT integration with predictive models supports automatic and data driven irrigation management.

Automating irrigation scheduling is a significant leap forward enabled by AIoT. AI-driven autonomous irrigation systems, which use ML, AI, and IoT sensors to automate irrigation decisions in real-time, have become revolutionary solutions in recent years [36–39]. AI models continuously analyze incoming sensor data to provide actionable insights or directly control irrigation equipment. These systems integrate IoT sensors, neural networks, and cloud connectivity to analyze real time data on soil moisture, temperature, and weather patterns [40]. AI models can forecast optimal irrigation conditions, automatically adjust water release valves, and maintain perfect soil moisture levels, resulting in reduced water uses and improved crop productivity [41]. Predictive irrigation scheduling uses ML techniques to forecast future irrigation needs rather than reacting only to present conditions. These models can identify indirect patterns and correlations that are not easily visible through conventional observation. Predictive models using ML techniques can enhance water use efficiency, crop yields, and agricultural sustainability [19,42]. These methods have demonstrated significant water savings and yield improvements compared to traditional scheduling techniques.

2.1.1. Deep learning-based crop water requirement forecasting

Deep learning (DL) models efficiently forecast crop water requirements using the complex interaction among soil, plant and climate variables. DL models such as Convolutional neural networks (CNN), Long Short-Term Memory (LSTM) and Hybrid models are used to predict evapotranspiration, soil moisture and irrigation need with precision [2, 43]. Integration of these models with AIoT provides real-time irrigation scheduling with optimum water use efficiency [42]. Deep learning, a subset of ML, can transform modern agriculture by allowing more precise and efficient crop growth monitoring, yield prediction, and early detection of plant diseases [44]. Deep learning models have shown significant potential in improving crop water requirement forecasting and irrigation management (Fig. 3). These models have emerged as effective solutions to overcome the limitations of traditional approaches in estimating crop water stress index, evapotranspiration and potential evapotranspiration [45]. Deep learning is being applied to various aspects of smart farming, including crop selection, yield prediction, and water management [46]. CNN and LSTM are commonly used for crop yield prediction, with a focus on cereals like wheat and corn [47]. These models can analyze complex datasets including satellite imagery, soil sensor readings, and weather data to predict crop water needs [48].

2.1.2. Reinforcement learning for autonomous irrigation optimization

Reinforcement learning (RL) provides a framework for autonomous irrigation management through system allows to learn optimal irrigation strategies using environmental variable interaction. Using the soil moisture sensing data, crop stress and weather variables, RL algorithms i.e. Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) adjust the irrigation timing and volume for optimum productivity when integrated with IoT sensing system [49]. RL is a ML learning framework that uses interaction with an environment, where an agent recognizes the state of that environment to develop a policy in terms of a specific objective [50]. In reinforcement learning-based irrigation systems, AI models test different irrigation strategies and receive feedback based on

Table 1

Overview of AIoT-based applications for agriculture management practices in precision agriculture.

Agriculture management practices	AIoT-based applications
Irrigation management	- Machine learning and AI models for predictive irrigation scheduling - Deep learning-based crop water requirement forecasting - Reinforcement learning for autonomous irrigation optimization - Digital twin models for simulating soil-water dynamics
Nutrient management	- AIoT for soil and crop nutrient monitoring and management - AI algorithms for effective fertilizer application
Disease management	- Plant diseases early detection through AIoT - AIoT based decision support system for disease management

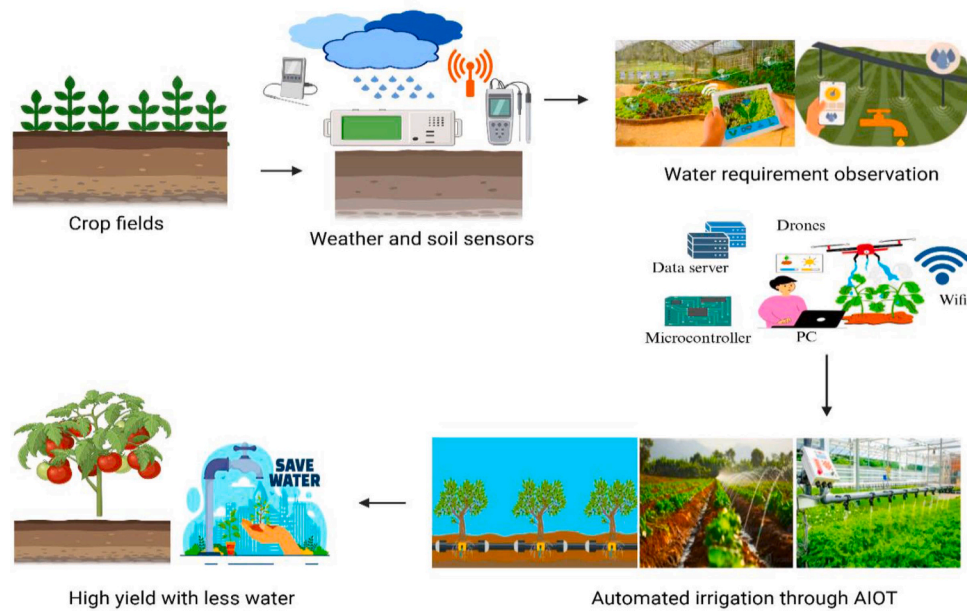


Fig. 2. AIoT-based smart technologies for precision irrigation management in agriculture.

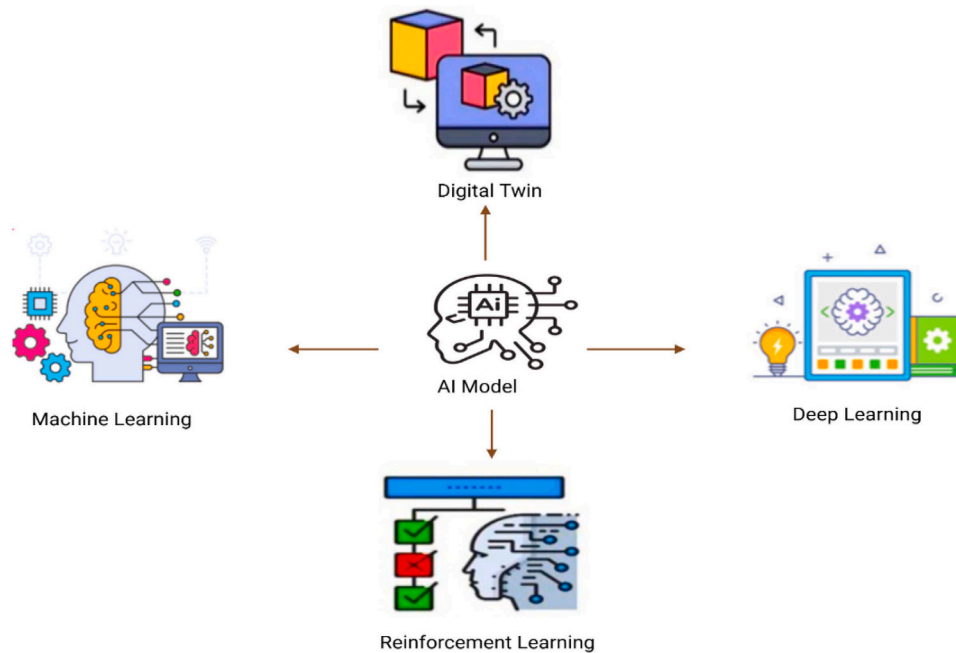


Fig. 3. AI-driven insights and models for automation of irrigation scheduling.

plant growth, yield, and water usage outcomes. Over time, the system learns the most efficient watering strategies for different crops under varying conditions. Deep reinforcement learning holds substantial potential for improving irrigation scheduling by adaptively determining water application based on active, time-series measurements across various cropping systems [51].

2.2. Digital twin models for simulating soil-water dynamics

A digital twin can integrate farm related information by leveraging technologies such as big data, IoT, and AI, and can alone activate operations based on system-generated decisions [52]. Digital twin models emerge as powerful tools for precision agriculture, offering virtual

representations of farms and crops to optimize resource management and decision-making [53]. Digital twin enables real-time communication between a physical component, such as an irrigation system, and its virtual counterpart in an IoT platform, where changes made in the physical system are automatically reflected in the digital model [54]. Digital twin model shows strong potential in agriculture, with current applications across crops, aquaponics and robotics, enabled by IoT, ML, and cyber-physical systems [55,56]. Digital twins are increasingly becoming important in irrigated agriculture, enabling smart water management [57]. For irrigation systems, a digital twin can monitor and control operations through sensors and actuators and several studies have demonstrated their use in combining climate, soil and crop sensor data to simulate irrigation strategies and improve water use efficiency

[58,59]. Digital twins thus bridge the gap between complex scientific models and practical field operations, making precision irrigation accessible to everyday farmers.

2.3. AIoT for irrigation system case studies

Advancements in AIoT technologies have altered traditional irrigation methods into intelligent, data-driven systems. The following case studies describe the real-world applications of these smart irrigation solutions. Gutierrez et al. [43], developed a solar-powered automated irrigation system using wireless sensor networks and general packet radio service for remote monitoring and control. Soil moisture and temperature sensors relay data via ZigBee-enabled units to a central controller, which communicates with a server for real-time irrigation management which saved up to 90 % water. Another study by Klein et al. [60], in USA demonstrated that spatially variable irrigation using a double drip line system and real time control can reduce vineyard variability and significantly improve yield and water use efficiency. Overall, a 26 % yield increase and 16 % water use efficiency improvement were achieved, proving the potential of digital data-driven irrigation. A practical example of AIoT implementation in precision

agriculture in India is the IoT-based intelligent irrigation module developed by Nawandar and Satpute, [61]. This low-cost system enables real-time monitoring of crop water needs and automates irrigation based on soil and plant conditions. It provides alerts to users, logs historical data, and supports efficient water use through intelligent decision-making, making it particularly suitable for greenhouses and small-scale farms. Karasekreter et al. [62], reported that integrating an artificial neural network model with soil physical properties and moisture content in a strawberry orchard resulted in energy savings of up to 23.9 % and water savings of 20.5 %. Dursun and Ozdan, [63], demonstrated the successful implementation of a wireless sensor-based drip irrigation system supported by solar energy and soil moisture sensors. The system optimized water use, reduced labor needs, and prevented moisture stress and soil salinization. Applied over 8 decares for cherry trees, it showcased the potential for real-time monitoring and automation in precision irrigation. Mehra et al. [64], proposed an innovative IoT-based hydroponic system that integrates deep neural networks to intelligently manage the growing environment. A smart IoT-based irrigation system was developed that integrates deep learning for automated plant recognition and water management [65]. The system uses image classification to identify plant types and a database to determine

Table 2

Overview of state-of-the-art AIoT approaches in smart irrigation with evaluation metrics and disadvantages.

Objective	Technology used	Evaluation metrics	Disadvantages	References
To develop an AIoT-based automated irrigation system to optimize water use efficiency.	Arduino ESP8266 microcontroller with Wi-Fi, TensorFlow Lite and ANN for irrigation decision-making, android studio app for visualization and control.	Model accuracy, precision, recall, F1 score and mean square error.	Requires reliable Wi-Fi connectivity between Arduino and smartphone, system complexity and sensor calibration needs, limited long-term rainfall prediction.	[17]
Develop and test an LSTM model for predicting long-term irrigation water requirements in vineyards using historical irrigation, soil moisture, and climate data.	ML based LSTM, data logger connected through cellular modem, wireless communication for remote access.	Mean squared error for predictive accuracy, standard deviation across folds, paired <i>t</i> -test for statistical significance.	Complex environmental variables not considered (crop phenology, soil texture), battery powered modem offline at night reducing data continuity.	[29]
To develop an AIoT-based precision irrigation system for sustainable water management and crop yield optimization.	AI algorithms such as random forest, decision tree regressor, gradient boost regressor, support vector regressor, and neural network regressor, cloud-IoT integration for real-time monitoring and prediction.	R-squared, mean absolute percentage error, comparative accuracy across AI models.	Requires strong internet connectivity and calibration of sensors hardware cost for Raspberry Pi and sensor network.	[31]
To propose a predictive algorithm-based irrigation management system that improves water use, prevents overwatering, and reduces energy consumption in tomato cultivation.	ML algorithms (KNN, ANN and SVM), LoRaWAN-based sensor network for real-time weather and soil data collection,	Accuracy and precision, recall and F1 score, confusion matrix, decision boundary visualization.	Limited validation on a single crop (tomato). requires high-quality datasets over full irrigation seasons.	[32]
Develop a smart IoT-based drip irrigation system that uses deep learning and adjusts water accordingly.	Deep learning with TensorFlow and ImageNet, Arduino.	Soil moisture response accuracy, flexibility and user convenience.	Not tested on large scale, accuracy may vary with image quality and training data.	[66]
Design a smart, autonomous drip irrigation system integrating IoT-based real-time sensing and LSTM based soil moisture prediction to reduce water waste, labor, and improve irrigation precision.	AIoT includes sensors and LSTM neural network.	Visual comparison of real and predicted soil moisture curves, irrigation trigger accuracy.	Requires reliable internet for remote control, hardware susceptibility to environmental noise.	[67]
Design and implement a smart farming system using EDGE-Fog-IoT-Cloud architecture to optimize irrigation and water usage in precision agriculture.	IoT wireless sensor network and ML,	Real-time data transmission, prediction accuracy for soil moisture, air temperature and humidity.	Training models are time consuming.	[68]
Develop a flexible, autonomous, data-driven irrigation system integrating IoT sensors and AI.	Soil moisture, temperature, humidity and rain sensor, LTSM deep learning model, Python framework for training.	Mean absolute error for prediction accuracy, training and validation loss analysis, responsiveness of irrigation control decisions.	Prototype limited to small-scale testing (three pots limited scalability validation), dependence on Wi-Fi connectivity for user commands although fallback logic exists, environmental conditions and crop types not generalized.	[69]
Develop an intelligent, low-cost irrigation prediction and automation approach using machine learning and IoT sensors.	Soil moisture, temperature, humidity and rain sensor, arduino microcontroller, neural network, support vector machine, logistic regression.	Root means square error to quantify prediction residuals, model comparison based on classification performance across ML algorithms.	Dataset limited to domestic-scale plants (not generalized to large fields), sensor data limited to a few environmental variables, requires continuous connectivity for cloud storage.	[70]

their water requirements, ensuring precise irrigation. By combining ML with soil moisture sensors, it adapts to watering based on real-time plant needs, enhancing water efficiency and minimizing human intervention. The overview of AIoT approaches in smart irrigation management is shown in Table 2.

3. AIoT for nutrient management

3.1. AIoT technology for precision nutrient management

AIoT is revolutionizing the way agriculture addresses the challenge of nutrient management. By merging the intelligent decision-making capabilities of AI with the sensing and communication power of IoT, AIoT enables data-driven, real-time, and precise nutrient applications. AIoT based smart fertilization strategies use field sensors to collect real-time data for AI-based algorithms, generating adaptive nutrient recommendations that increase yield and reduce environmental impact [28]. In the context of precision agriculture, nutrient management involves delivering the right amount of fertilizer to crops at the right time and location to maximize efficiency, yield, and environmental sustainability. AIoT enhances crop health monitoring and nutrient management, through remote sensing, imaging technologies, and predictive modeling [71].

3.2. AIoT for soil and crop nutrient monitoring

AIoT soil and crop nutrient monitoring involves deploying a range of IoT devices such as soil nutrient sensors, leaf color meters, and multi-spectral imaging systems to collect real-time data on critical parameters including soil nutrients, pH, moisture, organic matter, and crop nutrient deficiency indicators [37]. These systems transmit data wirelessly to cloud platforms for continuous analysis, enabling timely decision-making and precise soil management. ML algorithms analyze the collected data to recommend ideal crops and optimize fertilizer use, enhancing agricultural productivity and sustainability [72]. User-friendly mobile applications provide farmers with actionable insights and recommendations for nutrient improvement, promoting informed soil management practices and potentially increasing crop yields.

3.3. AI algorithms for effective fertilizer application

The core value of AI in nutrient management lies in its predictive and prescriptive capabilities. ML algorithms can forecast crop yields by analyzing fertilizer application rates, genetic traits, environmental conditions, and soil management practices [11]. ML algorithms, including support vector machines, decision trees, and neural networks, can process large datasets to develop predictive models for nutrient management [73]. These models use real time data to forecast optimal fertilizer dose, timing, and application method, empowering farmers with tailored, field-specific insights for more precise decision-making [74]. ML based Random Forest model provided fertilizer recommendations for rice, wheat, and maize, resulting in yield increases of approximately 1.3, 1.4, and 3.5 t ha⁻¹, respectively, while also enhancing net income and nutrient use efficiency [75]. Similarly, reinforcement learning algorithms represent a powerful data-driven method for improving farm management strategies, especially fertilizers application, ultimately enhancing nitrogen use efficiency [76]. Reinforcement learning algorithms is particularly effective because it can repeatedly adjust fertilization decisions based on feedback and contextual information from field conditions [77]. Collective learning techniques, such as stacking methods, have been employed to improve the accuracy of fertilizer recommendations by combining multiple ML models [78]. Overall, AI algorithms show promise in enhancing fertilizer use efficiency, improving crop yields, and reducing environmental impacts.

3.4. AIoT for nutrient management case studies

Several successful case studies have demonstrated the practical applications of AIoT in nutrient management. Ramakrishnam Raju, [79], developed AI-based smart hydroponic expert system (AI-SHES), integrating raspberry Pi, IoT, and a mobile app. It uses sensors and deep learning to monitor plant health and predict nutrient levels in real time. Senapaty et al. [80], proposed an IoT-SNA-CR (IoT-enabled soil nutrient classification and crop recommendation) model that employs IoT sensors and cloud-based data analysis to enable farmers to make informed decisions, reduce fertilizer use, and enhance productivity. Verma and Gawade, [81], proposed a ML framework to predict crop growth rate in hydroponic tomatoes using factors like dry weight, nutrient solution, ion uptake, EC limit, and nitrogen content. The study highlighted that Na⁺ ions had the highest positive correlation with dry weight, while Ca⁺ ions had the lowest. The model emphasizes the need for stage-specific ion monitoring for optimal tomato growth. Similarly, Rawade and Sahoo [82], developed a ML model to identify the most suitable crop type based on soil and atmospheric data. These case studies highlight the potential of AIoT to deliver tailored nutrient management solutions across various agricultural systems, contributing to increased nutrient use efficiency, reduced environmental impact, and improved farm profitability. Table 3 summarizes various case studies on AIoT application for nutrient management in detail.

4. AIoT for disease monitoring and management

4.1. AIoT for early disease detection

One of the most critical challenges in agriculture is the timely and accurate detection of plant diseases, as delayed identification often leads to severe yield losses. AIoT systems are revolutionizing crop disease management by integrating real-time environmental monitoring with ML models (Fig. 4). These models collect data on temperature, humidity, rainfall, and wind speed using IoT sensors and weather stations [83]. This data is then analyzed using various ML techniques, including support vector machines, logistic regression, and neural networks, to predict disease outbreaks. For instance, potato late blight can be forecasted by identifying specific combinations of temperature and humidity [84]. For plant disease detection, different AI-based approaches have been explored [85], contributing to the benefit of lowering pest pressure and farmers physical labor via remote monitoring, thereby lessening the need for traditional observation [86]. AI based systems using ML and computer vision enable early disease detection, supporting automated monitoring and decision making that optimize resource use and minimize agricultural losses [25]. Recent technological advancements have enabled automated methods for detecting wheat rust disease through ML and deep learning algorithms [87–91]. These AIoT systems not only enable early disease detection but also contribute to sustainable agriculture by optimizing pesticide use and improving crop management decisions.

4.2. AIoT based decision support system for disease management

Pests and diseases pose significant challenges, reducing both the yield and quality of crops [1]. AIoT-based decision support systems for disease management in agriculture have shown significant potential in improving crop health and reducing environmental impact. These systems utilize low cost IoT sensors to collect precise climate data, which is then analyzed using AI techniques to provide accurate predictions and recommendations for disease prevention [91]. Xenakis et al. [92], proposed a plant Disease Diagnosis Support System (DDSS) that combines an IoT-based platform and a lightweight agri-robotic system with AI algorithms to enable early plant disease detection with about 98 % accuracy. Cloud IoT frameworks enable wider accessibility and cost effectiveness compared to traditional methods [93]. These systems can

Table 3

Different case studies on AIoT applications in nutrient management for crops with objectives, technologies used and key evaluation metrics along with their limitations.

Objective	Technology used	Sensors/tools used	Parameters studied	Evaluation metrics	Disadvantages	References
To develop an ML enabled IoT system that continuously monitors soil nutrients, temperature, and moisture, and provides crop fertilizer recommendations.	IoT architecture with NodeMCU (ESP8266 Wi-Fi module), ML algorithms (CatBoost and Random Forest with grid SearchCV).	JXBS-3001 (NPK sensor), FC-28 (soil moisture sensor), DHT11 (temperature and humidity sensor), NodeMCU microcontroller.	Nutrient levels (NPK), Soil moisture, temperature and pH.	Accuracy and precision, model achieved accuracy of 97.5 % for crop and 99.56 % for fertilizer.	Sensitive to sensor calibration and data quality, requires reliable network connectivity, system difficulty and higher cost than conventional methods.	[11]
To develop a CNN-based and algorithm-driven prototype that optimizes fertilizer application using real-time field images, soil sensors, and environmental data.	CNN for soil image classification, algorithm integrating CNN output with IoT soil sensors and weather data,	Capacitive soil moisture sensor, JXCT (NPK sensor), DS18B20 (soil temperature sensor), ESP32/Arduino control board.	Soil texture, nutrient levels and moisture, air temperature, wind and rain forecast.	CNN classification accuracy (91.7 % real field and 97.1 % on test data), Validation accuracy/loss curves.	CNN trained only on loamy soil, limiting generalization to other soils, internet and power dependence.	[74]
Develop an AI-based smart hydroponic expert system (AI-SHES) integrating IoT and mobile applications for real-time monitoring, nutrient prediction, and disease detection in hydroponic farming.	Raspberry Pi controller, deep learning CNN, IoT cloud environment.	SHT-20 (Air temperature and humidity sensor), DS18B20 (water temperature sensor), SEN0161 (pH sensor), NPK sensor, hydrostatic pressure sensor.	Air temperature and humidity, water pH, nutrient levels (NPK, Mg, S, Ca).	Accuracy and precision, deep learning CNN attained 99.82 % accuracy for nutrient prediction.	High computational complexity, power demands in real-time environments and limited to hydroponic systems.	[79]
Develop an IoT-enabled soil nutrient analysis and crop recommendation model (IoT-SNA-CR) with real-time sensing, cloud storage, and ML-based classification	IoT, cloud storage, machine learning, GPS and Wi-Fi connectivity.	Soil, temperature and pH sensor.	Soil NPK, moisture, temperature and pH, water availability.	Model accuracy and precision.	High sensor installation and maintenance costs, requires farmer training for operation.	[80]
Design ML framework to predict absolute crop growth rate in hydroponic tomato crops using nutrient ion uptake, EC levels, dry weights, and nutrient solution data.	Machine learning algorithms.	Dionex DX500 for ion chromatography.	Plant dry weight, EC limit, nutrient solution consumption, ions uptake (Na, K, Ca, Mg, N).	Correlation analysis, ion uptake trends, crop growth rate trend following.	Lacks real-time sensing and automation, no direct sensor integration generally analytical and lab based.	[81]
Develop ML-based IoT system for predicting the utmost suitable crops based on soil and environmental data, using real-time sensors and past datasets.	IoT system, ML classification algorithms.	DHT22 (temperature and humidity sensor), EK1361 (soil moisture sensor), SEN0161 (pH sensor),	Soil temperature, humidity and pH, soil NPK values.	Accuracy and precision.	Complexity for rural farmers unaware of technology maintenance and cost worries.	[82]

estimate optimal fungicide quantities and application timing, leading to improved efficiency and reduced costs.

4.3. AIoT for disease monitoring case studies

Numerous case studies highlight the successful implementation of AIoT in disease monitoring and management. Karadağ et al. [94], developed a computerized diagnostic system to early detect plant diseases by analyzing light reflections from pepper leaves. Using ML algorithms such as ANN, Naive Bayes (NB) and K-nearest Neighbor (KNN), the system classified healthy, *Fusarium* infected, mycorrhizal infected, and combined infected leaves. The models achieved high accuracy such as 100 % with KNN, 97.5 % with ANN, and 90 % with NB, demonstrating strong potential for early disease detection in plants. Chen et al. [95], developed RiceTalk, an IoT and AI based platform for early detection of rice blast disease using non-image sensor data. Unlike traditional image-based methods, RiceTalk collects real-time environmental data from cultivation sensors, enabling timely disease prediction without manual data collection. The system integrates AI directly into the IoT network, reducing management costs and improving efficiency. Their novel spore germination-based feature extraction model achieved a prediction accuracy of 89.4 %. Another study by Mayalekshmi et al. [96], utilized the YOLOv5 model to detect leaf spot and leaf curl diseases in chili crops using real field images. The model achieved an accuracy of 75.64 % in identifying diseased cases from the test image dataset,

showing promise for practical disease diagnosis in agriculture. Arya and Singh, [97], explored the use of CNN and AlexNet for detecting diseases in potato and mango leaves. Using 4004 potato images from the Plant-Village database, AlexNet achieved a high accuracy of 98.33 %, proving its effectiveness in plant disease detection. Jiang et al. [98], developed a real-time apple leaf disease detection model using an improved CNN-based approach called INAR-SSD (SSD with an Inception module and Rainbow concatenation). Using a dataset of 26,377 images, the model accurately identified five common apple leaf diseases with a mean average precision of 78.80 % and a high detection speed of 23.13 frames per second. This method outperformed previous techniques in both accuracy and speed, offering a practical solution for early disease diagnosis in apple.

(Table 4)

5. The challenges of AIoT: adoption to precision farming

AIoT offers revolutionary potential for agriculture however, despite its promise, widespread adoption faces a variety of technical, economic, infrastructure, and ethical challenges that must be addressed to realize its full potential (Fig. 5). These barriers impact the scalability, accessibility, and reliability of AIoT systems, particularly in rural and resource-constrained regions where agriculture remains the backbone of livelihoods.

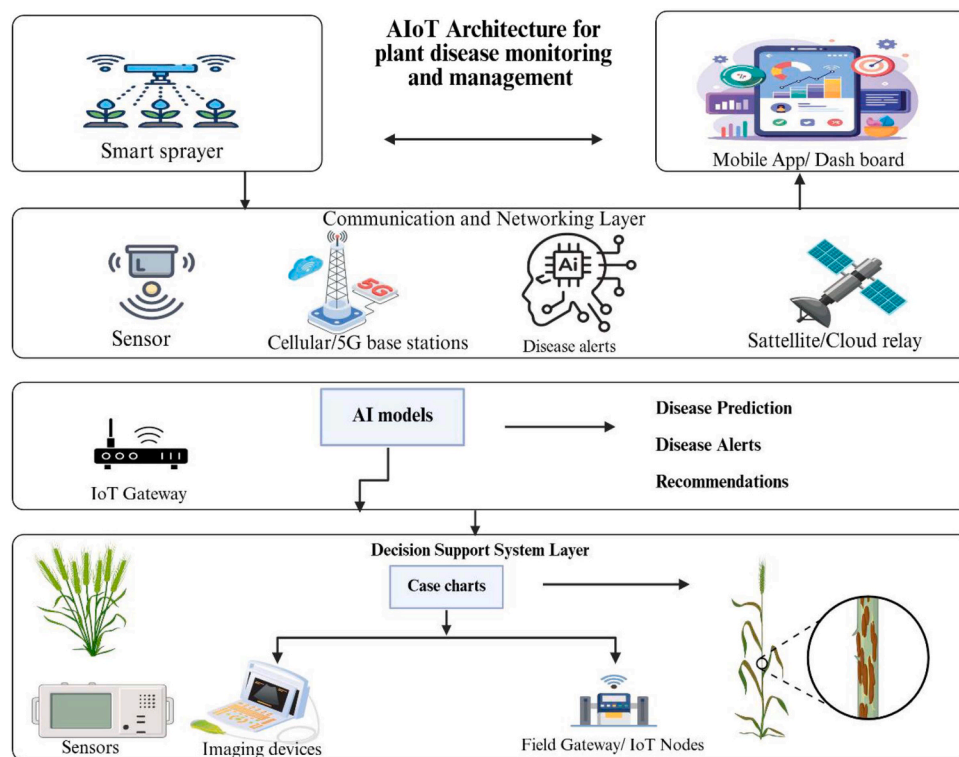


Fig. 4. AIoT-based framework for real-time plant disease monitoring and control: a visual overview.

Table 4

Different case studies on AIoT applications in plant disease monitoring with methodologies, technologies and outcomes.

Objective	Technology used	Dataset	Advantages	Limitations	Crop studied	References
Early detection of <i>Fusarium</i> disease in pepper plants using spectral reflectance (350–2500 nm) and classification via ML (KNN, ANN, NB).	Spectroradiometer (FieldSpec 3), ML algorithms (KNN, ANN, NB).	80 leaf samples from 4 groups (diseased/healthy $\pm$ mycorrhizal fungus), each with 2150 spectral data points.	Early disease detection before visual symptoms, non-destructive and cost-effective lab method, high accuracy (100 % KNN for binary classification).	High dimensional data (needed reduction), study limited to lab location not field-tested.	Pepper (<i>Capsicum annuum</i>)	[94]
To develop an AIoT system (RiceTalk) that detects rice blast disease using non-image-based sensors and AI, avoiding delays and manual efforts common with image-based systems.	IoT, CNN and cloud based dashboard.	5267 samples from 158 locations.	Real-time detection without human image capture, reduced labor/time cost spore germination used as a novel projecting feature, accuracy improved to 89.4 % in positive predictions.	Limited to environmental conditions influencing blast disease, barometric pressure and rain exposed minimal predictive effect, CNN model training requires careful tuning.	Rice (<i>Oryza sativa</i>)	[95]
To detect chilli crop leaf diseases (leaf spot and leaf curl) using YOLOv5 object detection model trained on in-field RGB images.	YOLOv5 (deep learning object detection model) Google Colab (GPU training).	210 in-field RGB images (160 training, 40 validation, 10 test).	Real-time detection capability, fast inference speed (13 ms/image), lightweight model (13.8 MB), trained on actual field data.	Small dataset size, leaf spot detection is less accurate due to non-uniform symptoms.	Chilli (<i>Capsicum annuum</i>)	[96]
Compare CNN and AlexNet deep learning architectures for classifying healthy and diseased mango and potato leaves using image data.	CNN and AlexNet (a pre-trained deep learning model).	Total 4004 images with 4 classes (2 healthy, 2 diseased).	AlexNet showed higher accuracy (98.33 %) than CNN (90.85 %), can be used on minimal hardware (CPU-based).	AlexNet takes more training time (50 min) than CNN (20 min).	Potato (<i>Solanum lycopersicum</i>), Mango (<i>Mangifera indica</i>)	[97]
Develop a real-time apple leaf disease detection model using an improved CNN (INAR-SSD), incorporating Inception modules and Rainbow concatenation for improved multi-scale feature removal.	Deep learning with INAR-SSD.	Total 26,377 images (from lab and real fields) include 5 disease classes.	High accuracy (78.80 %) and real-time speed (23.13 FPS), robust to small lesions and diverse image conditions, handles multi-disease detection in single images.	Some diseases like <i>Alternaria</i> and Grey spot are hard to differentiate due to alike features, model performance declines in low-light or noisy situations.	Apple (<i>Malus domestica</i>)	[98]

Challenges in Adopting AIoT in Precision Agriculture

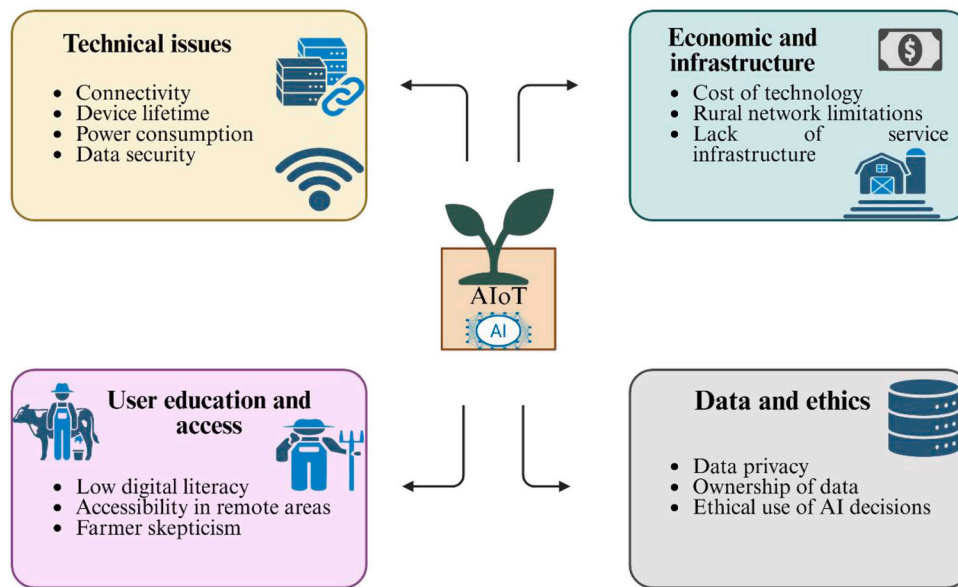


Fig. 5. Critical challenges hindering the adoption of AIoT technologies in the advancement of precision agriculture practices.

5.1. Technical issues

One of the most pressing technical challenges in deploying AIoT systems in agriculture is connectivity. Precision farming relies on continuous data exchange between sensors, cloud servers, and decision-support platforms. Conventional sensors rely on IoT components that transmit data from the soil and field to servers for processing and decision-making. However, their adoption is limited by certain factors such as cyber threats, high maintenance costs, and unreliable internet connectivity [99]. Many rural areas lack stable internet connectivity, hindering real-time data transmission crucial for precision farming [100]. In regions lacking networking infrastructure for backhaul, satellite IoT technologies provide a cost-effective solution for establishing connectivity [101]. Additionally, data security is a growing issue as farms become more digitized. Unprotected networks and unsecured devices are vulnerable to hacking, which could lead to the loss or manipulation of sensitive farm data and system disruptions.

5.2. Economic, infrastructural, educational and ethical challenges

Economic and infrastructure issues further hinder the adoption of AIoT in precision farming. High initial costs, lack of technical skills, and inadequate infrastructure are major barriers to implementation [102]. Even where financing is available, return on investment may not be immediate, making farmers uncertain to adopt unfamiliar technologies. Moreover, many countries lack physical infrastructure needed to support AIoT, such as reliable electricity, data storage facilities, or centralized farm data systems. For AIoT adoption to succeed, governments and private sectors must invest in infrastructure and provide incentives or financial support mechanisms tailored to diverse farming scales.

The lack of education and digital literacy among end users, particularly farmers, is another significant barrier. Many farmers are unfamiliar with AIoT tools or find them too complex to operate. Without proper training, even the most advanced systems can fail to produce results or be misused. Bridging this knowledge gap requires comprehensive capacity-building programs, including local language training, practical demonstrations, and access to extension services that integrate digital tools. In addition, gender and socioeconomic disparities can

affect access to these technologies. If these major challenges are not addressed, integrating IoT into agriculture could increase existing inequalities, allowing only the most well-resourced farmers to fully exploit the benefits of these technologies [103].

Finally, data protection and ethical concerns are increasingly relevant in the context of AIoT in agriculture. As farms become data-rich environments, questions arise about who owns the data, who has access to it, and how it is used. Many farmers are unwilling to share data due to worries of surveillance, exploitation by agribusiness corporations, or loss of control over their farm operations. To safeguard data privacy, it is important to educate farmers about data security [104,105]. Ethical fears also extend to algorithmic transparency farmers and stakeholders may not understand how AI models make decisions or why certain recommendations are made. This lack of transparency can reduce trust in AIoT systems. Establishing clear data governance frameworks, ensuring user consent, and creating open-source, farmer-centric platforms can help mitigate these concerns and foster responsible AIoT adoption.

6. Opportunities and future trends

6.1. AIoT technology advancement in agriculture

As AIoT technologies evolve, they bring forth intelligent, real time decision-making capabilities that surpass traditional farming methods. From predictive crop modeling to automated precision agricultural systems, AIoT allows for the development of smarter, more efficient agricultural practices. Advanced ML algorithms can analyze vast datasets collected by IoT devices such as weather stations, UAV, and soil sensors to forecast yields and recommend optimal input use.

AIoT capabilities and responsiveness in agriculture are expected to drastically enhance by the advent of 5 G and edge computing. The capacity of 5G networks has been greatly improved, enabling support for a vast number of IoT device connections [106]. Edge computing addresses latency issues by processing data closer to the source, reducing reliance on distant cloud servers. This combination enhances various applications, including smart farming, where it supports real-time monitoring, pest identification, and intelligent management [107]. At the same time, edge computing allows data to be processed locally on the device or near

the data source, minimizing reliance on cloud infrastructure and overcoming issues related to poor connectivity in remote areas. Table 5 illustrates the evaluation of key emerging technologies in detail.

6.2. AIoT for sustainable agriculture and climate change adaptation

AIoT plays a pivotal role in promoting sustainable agriculture and supporting climate change adaptation. By optimizing input usage such as water, fertilizers, and pesticides, AIoT can help reduce environmental degradation and reduce overall energy use [113]. AI-powered systems can anticipate weather extremes, droughts, or pest outbreaks, giving farmers the tools to proactively manage their fields and build climate resilience. Furthermore, adoption of AI technologies in agriculture contributes to quantifiable reductions in carbon footprints and improvements in overall energy efficiency [114]. As climate change continues to pose challenges to global food security, AIoT offers a transformative pathway to develop adaptive farming strategies and resilient agroecosystems. The adoption of these technologies through cost-effective solutions will be key to safeguarding that both large-scale and smallholder farmers can benefit from precision agriculture.

6.3. New applications, future directions and research gap

Looking forward, numerous new applications and future directions are emerging in the AIoT domain. Innovations such as autonomous tractors, AI-based crop breeding, blockchain-integrated supply chains, and AIoT enabled vertical farms are gaining momentum. The use of robotics for harvesting, AI for crop variety selection, and IoT enabled traceability systems for food safety are set to redefine agriculture from farm to fork. Future systems will likely integrate multi-source data from satellites, on-field sensors, social media, and economic markets to create highly intelligent, adaptive, and collaborative agricultural ecosystems.

AIoT in agriculture offers transformative gains in productivity and sustainability, but still adoption remains limited by key implementation barriers. Data standardization, interoperability, and cybersecurity remain persistent issues in scaling AIoT platforms. There is a need for extra farmer-centric AI models that are explainable, context-specific, and adaptive to diverse agroecological conditions. Research is also needed in areas like ethical AI use, reasonable technology distribution, and policy frameworks to regulate data governance in agriculture.

Table 5
Comparison of key emerging technologies and their pros and cons.

Technology	Pros	Cons	References
5G	• Ultra-fast data speeds • Low latency • Enabling emerging technologies	• Expensive infrastructure • Privacy and security risks	[108]
Edge AI	• Real-time processing • Reduced latency • Efficient bandwidth usage	• Limited computational power • Resource limitations • Security risks at edge devices	[109]
IoT (Internet of Things)	• Automation and efficiency • Data-rich insights	• Security vulnerabilities • Device interoperability challenges	[110]
Quantum computing	• Economic impact potential • Transformational computational power	• Lack of maturity (not yet ready for solving large-scale problems) • Error rates and scalability issues	[111]
Blockchain	• Transparent and protected • Decentralized trust model	• High energy consumption • High costs • Scalability and regulation problems	[112]

Moreover, developing low-cost AIoT solutions for smallholder farmers, particularly in developing countries, remains a critical challenge and opportunity. Collaborative efforts between academia, industry, and government will be essential to bridge these gaps and ensure that AIoT solutions are inclusive, sustainable, and transformative across agricultural systems.

7. Conclusion

The integration of AIoT into precision agriculture is revolutionizing conventional farming practices by enabling data-driven, real-time, and automated decision-making throughout different domains. This review highlights the pivotal role of AIoT technologies in optimizing irrigation, nutrient management, and disease control. Through the combination of sensors, edge computing, cloud platforms, and ML algorithms, AIoT enables farmers to achieve improved resource efficiency, enhanced crop productivity, and greater resilience against climate-induced stress. In the domain of smart irrigation, AIoT technologies facilitate dynamic water management by leveraging soil moisture sensors, weather stations, and predictive AI models. These technologies support the development of site-specific irrigation schedules, reduce water wastage, and encourage automated irrigation. Similarly, AIoT driven nutrient management systems integrate drone imagery, sensors, and AI-based recommendations to apply fertilizers precisely and efficiently, reducing input costs and minimizing environmental impact. Disease detection and management benefit from AIoT, the ability to monitor environmental conditions, recognize early signs of infection, and deliver timely alerts, thereby reducing crop damage and the overuse of pesticides. For farmers, AIoT aids as a powerful tool for gaining timely understandings into field conditions and making evidence-based decisions that enhance productivity and net income. However, the adoption of such technologies may be hindered by high initial investment costs, limited digital literacy, and connectivity constraints in rural areas. These challenges call for the development of user-friendly AIoT solutions and the implementation of supportive policies that promote accessibility and affordability. For agricultural scientists and engineers, the AIoT offers rich opportunities for interdisciplinary research in precision agriculture. Advancements in sensor design, AI, wireless networks, and data analytics can lead to the development of more robust and adaptive AIoT models tailored to diverse agroecological conditions. Furthermore, this field encourages collaboration between agronomists, data scientists, software developers, and environmental engineers to co-create solutions that bridge the gap between technological innovation and field level applications.

From a policy perspective, the successful distribution of AIoT in agriculture necessitates strong institutional support, particularly in terms of rural digital infrastructure, farmer training, and data governance. Governments must establish frameworks that address data privacy, cybersecurity, and ethical issues surrounding the use of AI in farming. Moreover, promoting public-private partnerships and farmer cooperatives can help scale up AIoT interventions and create shared value across the agricultural value chain. Looking ahead, AIoT is expected to be a cornerstone of future sustainable agriculture. Emerging technologies such as AI, IoT, 5G connectivity, edge computing, ML learning algorithms and digital twin modeling will further enhance the efficiency and intelligence of agricultural systems. These advancements will enable full integration of irrigation, nutrient, and disease management into a single, interoperable AIoT platform that can deliver complete and context-specific recommendations for crops. In summary, the combination of AI and IoT into the agricultural sector holds enormous potential to redefine farming practices. While certain technical, economic, and educational barriers persist, the long-term benefits of AIoT including enhanced productivity, efficient resource use, and environmental sustainability, make it a serious enabler of the next agricultural revolution. Joint efforts from scientists, policymakers, technology developers, and farming communities will be essential to fully harness the

transformative power of AIoT for precision agriculture.

Ethics statement not applicable

This manuscript does not include human or animal research.

CRediT authorship contribution statement

Jalal Bayar: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Nawab Ali:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Zhichao Cao:** Writing – review & editing, Methodology. **Yidong Ren:** Writing – review & editing, Methodology. **Younsuk Dong:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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